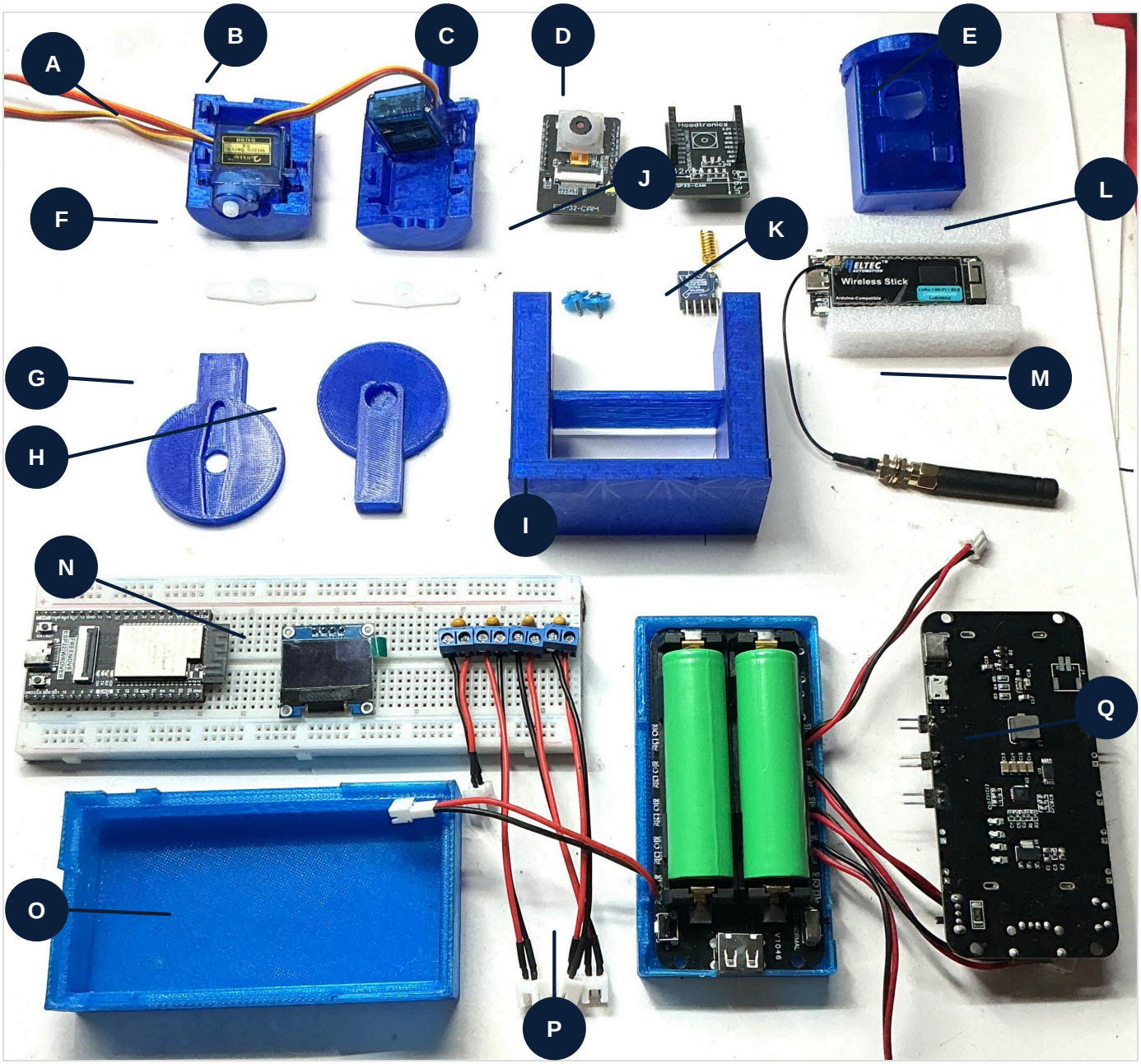
FIGURE 1A

CVisionBot (Dimension 1) — Component Layout, Pre-Assembly | Hoodtronics LLC | October 2026



Ref	Component	Notes
A	Pan servo — SG90/MG90S in 3D-printed blue gimbal mount	Horizontal pan axis actuator
B	Tilt servo — SG90/MG90S in 3D-printed blue gimbal mount	Vertical tilt axis actuator
C	AI Thinker ESP32-CAM module	Capture + Edge Impulse inference layer; TX on GPIO2
D	Hoodtronics ESP32-CAM custom breakout PCB	Proprietary interface board; integrates camera with LoRa UART lines

Ref	Component	Notes
E	CAM case-relay PCB enclosure / pan rotor — 3D-printed	Houses relay PCB; doubles as pan axis rotor
F	Servo horn spacers (white, ×2)	Constrain servo output shaft coupling to gimbal rotor
G	Tilt stator — 3D-printed, blue	Fixed frame element; anchors tilt axis
H	Tilt stator — 3D-printed, blue	Fixed frame element; anchors tilt axis (paired)
I	U-bracket base stand — 3D-printed, blue	Structural upright; supports pan-tilt column assembly
J	JST/Dupont header connector pins (blue, ×4)	Servo and UART wiring interface points
K	RYLR998 LoRa module with helical spring antenna	915 MHz LoRaWAN transceiver; AT command interface
L	Heltec Wireless Stick — ESP32 + SX1276 LoRa	Receiver node candidate; compact stick form factor
M	SMA stub antenna	External RF antenna for RYLR998 or Heltec Stick
N	Freenove ESP32-WROVER (no cam) on breadboard with OLED + terminal block	Transport layer node; GPIO16 RX from AI Thinker GPIO2 TX
O · P · Q — ESP32 Shield V3 Power Bank Assembly (one unit)		
O	3D-printed enclosure — top lid and bottom shell	Houses full power bank assembly; bottom shell sits under shield with battery; top lid closes the unit
P	ESP32 Shield V3 battery bottom — dual 18650 lithium cells	Underside layer of shield assembly; cells mount below the shield board; sits inside bottom enclosure shell (O)
Q	ESP32 Shield V3 board — solar charge controller, black PCB	Main shield board; sandwiched between battery bottom (P) and enclosure lid (O); solar-chargeable; USB output

Figure 1A. CVisionBot (Dimension 1) pre-assembly component layout showing pan servo (A), tilt servo (B), AI Thinker ESP32-CAM (C), Hoodtronics custom breakout PCB (D), enclosure cap (E), servo horn spacers (F), pan disc (G), tilt disc (H), U-bracket base (I), header pins (J), RYLR998 LoRa module (K), Heltec Wireless Stick (L), SMA antenna (M), Freenove ESP32-WROVER transport node on breadboard (N), and ESP32 Shield V3 power bank assembly: enclosure top/bottom shell (O), 18650 battery bottom layer (P), and Shield V3 board (Q) — O, P, and Q form one unit. All 3D-printed components fabricated by Hoodtronics LLC.